

CBSE Class-12 Mathematics

NCERT Exemplar Solutions

Chapter - 10 Vector Algebra

Solved Examples

Short Answer (S.A.)

1. Find the unit vector in the direction of the sum of the vectors $\vec{a}=2\hat{i}-\hat{j}+2\hat{k}$ and $\vec{b}=-\hat{i}+\hat{j}+3\hat{k}$.

Ans: Let $\vec{c}$ be the sum of given vectors i.e $\vec{a}$ and $\vec{b}$

$$\vec{c} = \vec{a} + \vec{b}$$

$$\vec{c} = 2\hat{i} - \hat{j} + 2\hat{k} - \hat{i} + \hat{j} + 3\hat{k}$$

$$\vec{c} = \hat{i} + 5\hat{k}$$

$$|\vec{c}| = \sqrt{1^2 + 5^2} = \sqrt{26}$$

Now, the unit vector is given by

$$\hat{c} = \frac{\vec{c}}{|\vec{c}|} = \frac{\hat{i} + 5\hat{k}}{\sqrt{26}} = \frac{1}{\sqrt{26}}\hat{i} + \frac{5}{\sqrt{26}}\hat{k}$$

2. Find a vector of magnitude 11 in the direction opposite to that of $\overrightarrow{PQ}$, where P and Q are the points (1, 3, 2) and (-1, 0, 8), respectively.

Ans: The vector with initial point P (1, 3, 2) and terminal point Q (-1, 0, 8) is given by

$$\overrightarrow{PQ} = (-1-1)\hat{i} + (0-3)\hat{j} + (8-2)\hat{k}$$

$$\overrightarrow{PQ} = -2\hat{i} - 3\hat{j} + 6\hat{k}$$

We know, $\overrightarrow{QP} = -\overrightarrow{PQ}$

$$\overrightarrow{QP} = -(-2\hat{i} - 3\hat{j} + 6\hat{k})$$

$$\overrightarrow{QP} = 2\hat{i} + 3\hat{j} - 6\hat{k}$$

Now, the unit vector in the direction of $\overrightarrow{QP}$ is given by

$$\widehat{QP} = \frac{\overrightarrow{QP}}{|\overrightarrow{QP}|} = \frac{2\hat{i} + 3\hat{j} - 6\hat{k}}{\sqrt{2^2 + 3^2 + (-6)^2}} = \frac{2\hat{i} + 3\hat{j} - 6\hat{k}}{\sqrt{49}} = \frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} - \frac{6}{7}\hat{k}$$

Therefore, the vector of magnitude 11 in direction of $\overrightarrow{QP}$ is given by

$$11\widehat{QP} = 11\left(\frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} - \frac{6}{7}\hat{k}\right) = \frac{22}{7}\hat{i} + \frac{33}{7}\hat{j} - \frac{66}{7}\hat{k}$$

3. Find the position vector of a point R which divides the line joining the two points P and Q with position vectors $\overrightarrow{OP} = 2\hat{a} + \hat{b}$ and $\overrightarrow{OQ} = \hat{a} - 2\hat{b}$, respectively, in the ratio 1:2, (i) internally and (ii) externally.

Ans: (i) The position vector of the point R which is dividing the join of P and Q internally in the ratio 1:2 is given by

$$\overrightarrow{OR} = \frac{2(2\hat{a} + \hat{b}) + 1(\hat{a} - 2\hat{b})}{1 + 2} = \frac{5\hat{a}}{3}$$

(ii) The position vector of the point R' which is dividing the join of P and Q in the ratio 1 : 2 externally is given by

$$\overrightarrow{OR'} = \frac{2(2\hat{a} + \hat{b}) - 1(\hat{a} - 2\hat{b})}{2 - 1} = 3\hat{a} + 4\hat{b}$$

4. If the points $(-1, -1, 2)$, $(2, m, 5)$ and $(3, 11, 6)$ are collinear, find the value of m.

Ans: Given points are : A $(-1, -1, 2)$, B $(2, m, 5)$ and C $(3, 11, 6)$.

It is given that the points are collinear

$$\overrightarrow{AB} = (2+1)\hat{i} + (m+1)\hat{j} + (5-2)\hat{k}$$

$$\overrightarrow{AB} = 3\hat{i} + (m+1)\hat{j} + 3\hat{k}$$

Also,

$$\overrightarrow{AC} = (3+1)\hat{i} + (11+1)\hat{j} + (6-2)\hat{k}$$

$$\overrightarrow{AC} = 4\hat{i} + 12\hat{j} + 4\hat{k}$$

Since A, B, C, are collinear, we have

$$\overrightarrow{AB} = \lambda \overrightarrow{AC}$$

$$3\hat{i} + (m+1)\hat{j} + 3\hat{k} = \lambda(4\hat{i} + 12\hat{j} + 4\hat{k})$$

$$\Rightarrow 3 = 4\lambda \text{ and } m+1 = 12\lambda$$

Hence, $m = 8$

5. Find a vector $\vec{r}$ of magnitude $3\sqrt{2}$ units which makes an angle of $\frac{\pi}{4}$ and $\frac{\pi}{2}$ with y and z - axes, respectively.

Ans: Here we have been given m and n as

$$m = \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\text{and } n = \cos \frac{\pi}{2} = 0$$

we know that

$$l^2 + m^2 + n^2 = 1$$

$$l^2 + \frac{1}{2} + 0 = 1$$

$$l = \pm \frac{1}{\sqrt{2}}$$

Therefore, the required vector $\vec{r} = 3\sqrt{2} (l\hat{i} + m\hat{j} + n\hat{k})$

$$\vec{r} = 3\sqrt{2} \left(\pm \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + 0\hat{k} \right)$$

$$\vec{r} = \pm 3\hat{i} + 3\hat{j}$$

Hence, $\vec{r} = \pm 3\hat{i} + 3\hat{j}$

6. If $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} - 2\hat{k}$ and $\vec{c} = \hat{i} + 3\hat{j} - \hat{k}$, find λ such that $\vec{a}$ is perpendicular to $\lambda\vec{b} + \vec{c}$.

Ans: We have

$$\lambda\vec{b} + \vec{c} = \lambda(\hat{i} + \hat{j} - 2\hat{k}) + (\hat{i} + 3\hat{j} - \hat{k})$$

$$\lambda\vec{b} + \vec{c} = (\lambda + 1)\hat{i} + (\lambda + 3)\hat{j} - (2\lambda + 1)\hat{k}$$

Now since $\vec{a}$ is perpendicular to $\lambda\vec{b} + \vec{c}$

$$\vec{a} \cdot (\lambda\vec{b} + \vec{c}) = 0$$

$$(2\hat{i} - \hat{j} + \hat{k}) \cdot (\lambda\vec{b} + \vec{c}) = 0$$

$$(2\hat{i} - \hat{j} + \hat{k}) \cdot [(\lambda + 1)\hat{i} + (\lambda + 3)\hat{j} - (2\lambda + 1)\hat{k}] = 0$$

$$2(\lambda + 1) - (\lambda + 3) - (2\lambda + 1) = 0$$

$$\lambda = -2$$

Hence the value of $\lambda = -2$

7. Find all vectors of magnitude $10\sqrt{3}$ that are perpendicular to the plane of $\hat{i} + 2\hat{j} + \hat{k}$ and $-\hat{i} + 3\hat{j} + 4\hat{k}$.

Ans: Let us assume that $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$ and $\vec{b} = -\hat{i} + 3\hat{j} + 4\hat{k}$

Now , $\vec{a} \times \vec{b} = (\hat{i} + 2\hat{j} + \hat{k}) \times (-\hat{i} + 3\hat{j} + 4\hat{k})$

$$\vec{a} \times \vec{b} = (\hat{i} + 2\hat{j} + \hat{k}) \times (-\hat{i} + 3\hat{j} + 4\hat{k})$$

$$\vec{a} \times \vec{b} = \hat{i}(8-3) - \hat{j}(4+1) + \hat{k}(3+2)$$

$$\vec{a} \times \vec{b} = 5\hat{i} - 5\hat{j} + 5\hat{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{5^2 + (-5)^2 + 5^2} = 5\sqrt{3}$$

Unit vector perpendicular to the plane of a and b is given by

$$\frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{5\hat{i} - 5\hat{j} + 5\hat{k}}{5\sqrt{3}}$$

Therefore , vectors of magnitude of $10\sqrt{3}$ that are perpendicular to plane of a and b are $\pm 10\sqrt{3} \left(\frac{5\hat{i} - 5\hat{j} + 5\hat{k}}{5\sqrt{3}} \right) = \pm 10(\hat{i} - \hat{j} + \hat{k})$

Long Answer (L.A)

8. Using vectors, prove that $\cos(A - B) = \cos A \cos B + \sin A \sin B$.

Ans:

Let $\widehat{OP}$ and $\widehat{OQ}$ be unit vectors making angles A and B, respectively, with positive direction of x-axis. Then $\angle QOP = A - B$

Now , We know $\widehat{OP} = \widehat{OM} + \widehat{MP} = \hat{i}\cos A + \hat{j}\sin A$

and $\widehat{OQ} = \widehat{ON} + \widehat{NQ} = \hat{i}\cos B + \hat{j}\sin B$

By definition , we have

$$\widehat{OP} \cdot \widehat{OQ} = |\widehat{OP}| |\widehat{OQ}| \cos(A - B) = \cos(A - B) \dots \dots \dots \text{eqn (1)}$$

In terms of components, we have

$$\widehat{OP} \cdot \widehat{OQ} = (\hat{i}\cos A + \hat{j}\sin A) \cdot (\hat{i}\cos B + \hat{j}\sin B)$$

$$= \cos A \cos B + \sin A \sin B \quad \dots\dots\dots \text{eqn(2)}$$

From (1) and (2), we have

$$\cos (A - B) = \cos A \cos B + \sin A \sin B.$$

Hence Proved

9. Prove that in a ΔABC , $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$ where a, b, c represent the magnitudes of the sides opposite to vertices A, B, C, respectively.

Ans: Let the three sides of the triangle BC, CA and AB be represented by $\vec{a}$, $\vec{b}$, and $\vec{c}$ respectively

Now we have ,

$$\vec{a} + \vec{b} + \vec{c} = 0$$

$$\Rightarrow \vec{a} + \vec{b} = -\vec{c}$$

On pre cross multiplying by $\vec{a}$, and post cross multiplying by $\vec{b}$, we get

$$\vec{a} \times \vec{b} = \vec{c} \times \vec{a}$$

$$\text{and } \vec{a} \times \vec{b} = \vec{b} \times \vec{c}$$

$$\text{Therefore , } \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

$$|\vec{a} \times \vec{b}| = |\vec{b} \times \vec{c}| = |\vec{c} \times \vec{a}|$$

$$|\vec{a}| |\vec{b}| \sin (\pi - C) = |\vec{b}| |\vec{c}| \sin (\pi - A) = |\vec{c}| |\vec{a}| \sin (\pi - B)$$

$$ab \sin C = bc \sin A = ca \sin B$$

Dividing by abc, we get

$$\frac{\sin C}{c} = \frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Hence Proved

Objective Type Questions

10. The magnitude of the vector $6\hat{i}+2\hat{j}+3\hat{k}$ is

- (A) 5
- (B) 7
- (C) 12
- (D) 1

Ans: Let $\vec{a} = 6\hat{i}+2\hat{j}+3\hat{k}$

Now magnitude of $\vec{a}$ is given by

$$|\vec{a}| = \sqrt{6^2 + 2^2 + 3^2} = \sqrt{36+4+9} = \sqrt{49} = 7$$

Therefore magnitude of the vector $\vec{a}$ is 7units

Hence the correct option is (B)

11.The position vector of the point which divides the join of points with position vectors $\vec{a} + \vec{b}$ and $2\vec{a} - \vec{b}$ in the ratio 1 : 2 is

- (A) $\frac{3\vec{a}+2\vec{b}}{3}$
- (B) $\vec{a}$
- (C) $\frac{5\vec{a}-\vec{b}}{3}$
- (D) $\frac{4\vec{a}+\vec{b}}{3}$

Ans: The position vector of required point by using section formula is calculated as

$$\frac{2(\vec{a}+\vec{b})+1(2\vec{a}-\vec{b})}{2+1} = \frac{4\vec{a}+\vec{b}}{3}$$

Hence the correct option is (D)

12. The vector with initial point P (2, -3, 5) and terminal point Q(3, -4, 7) is

(A) $\hat{i} - \hat{j} + 2\hat{k}$

(B) $5\hat{i} - 7\hat{j} + 12\hat{k}$

(C) $-\hat{i} + \hat{j} - 2\hat{k}$

(D) 0

Ans: Initial point P (2, -3, 5)

Terminal point Q(3, -4, 7)

The vector is given by

$$(3-2)\hat{i} + (-4-(-3))\hat{j} + (7-5)\hat{k} = \hat{i} - \hat{j} + 2\hat{k}$$

Hence the correct option is (A)

13. The angle between the vectors $\hat{i} - \hat{j}$ and $\hat{j} - \hat{k}$ is

(A) $\frac{\pi}{3}$

(B) $\frac{2\pi}{3}$

(C) $\frac{-\pi}{3}$

(D) $\frac{5\pi}{6}$

Ans: The angle between two vectors is given by

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{(\hat{i} - \hat{j}) \cdot (\hat{j} - \hat{k})}{|\hat{i} - \hat{j}| \cdot |\hat{j} - \hat{k}|} = \frac{-1}{\sqrt{1^2 + 1^2} \cdot \sqrt{1^2 + 1^2}} = \frac{-1}{2}$$

$$\theta = \frac{2\pi}{3}$$

Hence the correct option is (B)

14. The value of λ for which the two vectors $2\hat{i} - \hat{j} + 2\hat{k}$ and $3\hat{i} + \lambda\hat{j} + \hat{k}$ are perpendicular is

(A) 2

(B) 4

(C) 6

(D) 8

Ans: Since the vectors are perpendicular to each other

$$(2\hat{i} - \hat{j} + 2\hat{k}) \cdot (3\hat{i} + \lambda\hat{j} + \hat{k}) = 0$$

$$6 - \lambda + 2 = 0$$

$$\lambda = 8$$

Hence the correct option is (D)

15. The area of the parallelogram whose adjacent sides are $\hat{i} + \hat{k}$ and $2\hat{i} + \hat{j} + \hat{k}$ is

(A) $\sqrt{2}$

(B) $\sqrt{3}$

(C) 3

(D) 4

Ans: Area of parallelogram whose adjacent sides are $\vec{a}$ and $\vec{b}$ is given by $|\vec{a} \times \vec{b}|$

$$\vec{a} \times \vec{b} = (\hat{i} + \hat{k}) \times (\widehat{2i} + \hat{j} + \hat{k})$$

$$\vec{a} \times \vec{b} = -\hat{i} + \hat{j} + \hat{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{(-1)^2 + (1)^2 + (1)^2} = \sqrt{3}$$

Hence the correct option is (B)

16. If $|\vec{a}| = 8$, $|\vec{b}| = 3$ and $|\vec{a} \times \vec{b}| = 12$, then value of $\vec{a} \cdot \vec{b}$ is

(A) $6\sqrt{3}$

(B) $8\sqrt{3}$

(C) $12\sqrt{3}$

(D) None of these

Ans: we know that

$$|\vec{a} \times \vec{b}| = |\vec{a}| \cdot |\vec{b}| |\sin \theta|$$

$$12 = 8 \cdot 3 \cdot |\sin \theta|$$

$$|\sin \theta| = \frac{1}{2}$$

$$\theta = \pm \frac{\pi}{6}$$

Now we have,

$$\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| |\cos \theta|$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| |\cos \theta| = 8 \times 3 \times \frac{\sqrt{3}}{2} = 12\sqrt{3}$$

Therefore, $\vec{a} \cdot \vec{b} = 12\sqrt{3}$

Hence the correct option is (C)

17, The 2 vectors $\hat{j}+\hat{k}$ and $3\hat{i}-\hat{j}+4\hat{k}$ represents the two sides AB and AC, respectively of a ΔABC . The length of the median through A is

(A) $\frac{\sqrt{34}}{2}$

(B) $\frac{\sqrt{48}}{2}$

(C) $\sqrt{18}$

(D) None of these

Ans: We know that Median $\overline{AD}$ is given by

$$|\overline{AD}| = \frac{1}{2} |3\hat{i} - \hat{j} + 4\hat{k}|$$

$$|\overline{AD}| = \frac{\sqrt{34}}{2}$$

Hence the correct option is (A)

18. The projection of vector $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ along $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$ is

(A) $\frac{2}{3}$

(B) $\frac{1}{3}$

(C) 2

(D) $\sqrt{6}$

Ans: Projection of a vector $\vec{a}$ on $\vec{b}$ is given by

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{(2\hat{i} - \hat{j} + \hat{k}) \cdot (\hat{i} + 2\hat{j} + 2\hat{k})}{(|2\hat{i} - \hat{j} + \hat{k}|)(|\hat{i} + 2\hat{j} + 2\hat{k}|)} = \frac{2}{3}$$

Hence the correct option is (A)

19. If $\vec{a}$ and $\vec{b}$ are unit vectors, then what is the angle between $\vec{a}$ and $\vec{b}$ for $\sqrt{3}\vec{a} - \vec{b}$ to be a unit vector?

(A) 30°

(B) 45°

(C) 60°

(D) 90°

Ans: Here we have

$$(\sqrt{3}\vec{a} - \vec{b})^2 = 3(\vec{a})^2 + (\vec{b})^2 - 2\sqrt{3}\vec{a} \cdot \vec{b}$$

$$\vec{a} \cdot \vec{b} = \frac{\sqrt{3}}{2}$$

$$\cos\theta = \frac{\sqrt{3}}{2}$$

$$\theta = 30^\circ.$$

Hence the correct option is (A)

20. The unit vector perpendicular to the vectors $\hat{i} - \hat{j}$ and $\hat{i} + \hat{j}$ forming a right handed system is

(A) $\hat{k}$

(B) $-\hat{k}$

(C) $\frac{\hat{i} - \hat{j}}{\sqrt{2}}$

(D) $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$

Ans: Unit vector perpendicular to the vectors $\hat{i} - \hat{j}$ and $\hat{i} + \hat{j}$ is given by

$$\frac{(\hat{i} - \hat{j}) \cdot (\hat{i} + \hat{j})}{|(\hat{i} - \hat{j}) \cdot (\hat{i} + \hat{j})|} = \hat{k}$$

Hence the correct option is (A)

21. If $|\vec{a}| = 3$ and $-1 \leq k \leq 2$, then $|k\vec{a}|$ lies in the interval

(A) $[0, 6]$

(B) $[-3, 6]$

(C) $[3, 6]$

(D) $[1, 2]$

Ans: Given $|\vec{a}| = 3$ and $-1 \leq k \leq 2$,

The smallest value of $|k\vec{a}|$ will exist at the numerically smallest value of k , i.e., at $k = 0$, which gives

$$|k\vec{a}| = |k| |\vec{a}| = 0 \times 3 = 0$$

The numerically greatest value of k is 2 at which $|k\vec{a}| = 6$.

$$|k\vec{a}| \in [0, 6]$$

Hence the correct option is (A)

Exercise – 10.3

Short Answer (S.A.)

1. Find the unit vector in the direction of sum of vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{j} + \hat{k}$.

Ans:

Let $\vec{c}$ denotes the sum of $\vec{a}$ and $\vec{b}$. So,

$$\vec{c} = \vec{a} + \vec{b} = 2\hat{i} - \hat{j} + \hat{k} + (2\hat{j} + \hat{k}) = 2\hat{i} + \hat{j} + 2\hat{k}$$

$$|\vec{c}| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{9} = 3$$

Unit vector along vector c will be

$$\hat{c} = \frac{\vec{c}}{|\vec{c}|} = \frac{2\hat{i} + \hat{j} + 2\hat{k}}{3} = \frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k}$$

2. If $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - 2\hat{k}$ find the unit vector in the direction of

(i) $6\vec{b}$

(ii) $2\vec{a} - \vec{b}$

Ans:

Given that $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - 2\hat{k}$

(i) We have to find unit vector along

$$6\vec{b} = 12\hat{i} + 6\hat{j} - 12\hat{k}$$

We know that all parallel vectors have same unit vector so, unit vector along $6\vec{b}$ will be same as along $\vec{b}$

$$\hat{b} = \frac{\vec{b}}{|\vec{b}|} = \frac{2\hat{i} + \hat{j} - 2\hat{k}}{\sqrt{2^2 + 1^2 + 2^2}} = \frac{1}{\sqrt{9}}(2\hat{i} + \hat{j} - 2\hat{k})$$

$$\hat{b} = \frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$$

(ii) We have to find unit vector along $\vec{c} = 2\vec{a} - \vec{b}$

$$\vec{c} = 2\vec{a} - \vec{b} = 2(\hat{i} + \hat{j} + 2\hat{k}) - (2\hat{i} + \hat{j} - 2\hat{k})$$

$$= \hat{j} + 6\hat{k}$$

$$\hat{c} = \frac{\vec{c}}{|\vec{c}|} = \frac{\hat{j} + 6\hat{k}}{\sqrt{1^2 + 6^2}} = \frac{1}{\sqrt{37}}(\hat{j} + 6\hat{k})$$

3. Find a unit vector in the direction of $\overrightarrow{PQ}$ where P and Q have coordinates (5, 0, 8) and (3, 3, 2), respectively.

Ans:

Given that P and Q have coordinates (5, 0, 8) and (3, 3, 2).

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = (3\hat{i} + 3\hat{j} + 2\hat{k}) - (5\hat{i} + 8\hat{k}) = -2\hat{i} + 3\hat{j} - 6\hat{k}$$

$$\widehat{\overrightarrow{PQ}} = \frac{\overrightarrow{PQ}}{|\overrightarrow{PQ}|} = \frac{-2\hat{i} + 3\hat{j} - 6\hat{k}}{\sqrt{2^2 + 3^2 + 6^2}} = \frac{1}{\sqrt{49}}(-2\hat{i} + 3\hat{j} - 6\hat{k})$$

$$\widehat{\overrightarrow{PQ}} = \frac{1}{7}(-2\hat{i} + 3\hat{j} - 6\hat{k})$$

4. If $\vec{a}$ and $\vec{b}$ are the position vectors of A and B, respectively, find the position vector of a point C in BA produced such that $BC = 1.5 BA$.

Ans:

Given that $OA = \vec{a}$ and $OB = \vec{b}$

$$BA = OA - OB = \vec{a} - \vec{b}$$

$$BC = 1.5BA = 1.5(\vec{a} - \vec{b}).$$

$$BC = OC - OB = \vec{c} - \vec{b}$$

$$1.5(\vec{a} - \vec{b}) = \vec{c} - \vec{b}$$

$$\vec{c} = 1.5\vec{a} - 1.5\vec{b} + \vec{b}$$

$$OC = \vec{c} = \frac{1}{2}(3\vec{a} - \vec{b})$$

5. Using vectors, find the value of k such that the points A (k, -10, 3), B (1, -1, 3) and C (3, 5, 3) are collinear.

Ans:

Given points are $A(k, -10, 3)$, $B(1, -1, 3)$ and $C(3, 5, 3)$ Since points are collinear so

$$AB = x(BC) \Rightarrow OB - OA = x(OC - OB)$$

$$(\hat{i} - \hat{j} + 3\hat{k}) - (k\hat{i} - 10\hat{j} + 3\hat{k}) = x[(3\hat{i} + 5\hat{j} + 3\hat{k}) - (\hat{i} - \hat{j} + 3\hat{k})]$$

$$(1-k)\hat{i} + 9\hat{j} = x(2\hat{i} + 6\hat{j})$$

By comparing the terms, we have

$$1-k = 2x \Rightarrow k = 1-2x$$

$$9 = 6x \Rightarrow 2x = 3$$

$$k = 1-3 = -2$$

Hence value of $k = -2$.

6. A vector $\vec{r}$ is inclined at equal angles to the three axes. If the magnitude of $\vec{r}$ is $2\sqrt{3}$ units, find $\vec{r}$.

Ans:

Given $|\vec{r}| = 2\sqrt{3}$ and $\vec{r}$ is equally inclined to axes so

$$l = m = n$$

$$l^2 + m^2 + n^2 = 1 \Rightarrow 3l^2 = 1$$

$$l = \pm \frac{1}{\sqrt{3}} = m = n$$

$$\text{Now, } \vec{r} = |\vec{r}|(l\hat{i} + m\hat{j} + n\hat{k})$$

$$\vec{r} = 2\sqrt{3}\left(\pm \frac{1}{\sqrt{3}}\hat{i} \pm \frac{1}{\sqrt{3}}\hat{j} \pm \frac{1}{\sqrt{3}}\hat{k}\right) = \pm 2(\hat{i} + \hat{j} + \hat{k})$$

7. A vector $\vec{r}$ has magnitude 14 and direction ratios 2, 3, -6. Find the direction cosines and components of $\vec{r}$, given that $\vec{r}$ makes an acute angle with x-axis.

Ans:

Given that $|\vec{r}| = 14$ and direction ratios 2, 3, -6.

Let direction cosines are l, m and n

$$\therefore l = \frac{2x}{14} = \frac{x}{7}$$

$$\therefore m = \frac{3x}{14}$$

$$\therefore n = -\frac{6x}{14} = -\frac{3x}{7}$$

$$\text{Now, } l^2 + m^2 + n^2 = 1$$

$$\left(\frac{x}{7}\right)^2 + \left(\frac{3x}{14}\right)^2 + \left(-\frac{3x}{7}\right)^2 = 1$$

$$\frac{x^2}{49} + \frac{9x^2}{196} + \frac{9x^2}{49} = 1$$

$$(4 + 9 + 36)x^2 = 196$$

$$49x^2 = 4 \times 49$$

$$x^2 = 4$$

$$\therefore x = \pm 2$$

Since $\vec{r}$ makes acute angle with x-axis so value of $x = 2$

$$\text{Now, } l = \frac{2}{7}, m = \frac{3}{7} \text{ and } n = -\frac{6}{7}$$

$$\vec{r} = |\vec{r}|(\hat{i}l + \hat{j}m + \hat{k}n) = 14\left(\frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} - \frac{6}{7}\hat{k}\right)$$

$$\vec{r} = 4\hat{i} + 6\hat{j} - 12\hat{k}$$

8. Find a vector of magnitude 6, which is perpendicular to both the vectors $2\hat{i}-\hat{j}+2\hat{k}$ and $4\hat{i}-\hat{j}+3\hat{k}$.

Ans:

Let $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = 4\hat{i} - \hat{j} + 3\hat{k}$

Vector product of the vectors is always perpendicular to both the vectors. So,

$$\vec{c} = \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 2 \\ 4 & -1 & 3 \end{vmatrix} = (-3+2)\hat{i} - (6-8)\hat{j} + (-2+4)\hat{k}$$

$$\vec{c} = -\hat{i} + 2\hat{j} + 2\hat{k}$$

Required vector should have magnitude 6 then

$$\vec{r} = 6 \times \frac{\vec{c}}{|\vec{c}|} = 6 \times \frac{-\hat{i} + 2\hat{j} + 2\hat{k}}{\sqrt{1^2 + 2^2 + 2^2}} = \frac{6}{\sqrt{9}}(-\hat{i} + 2\hat{j} + 2\hat{k})$$

$$\vec{r} = -2\hat{i} + 4\hat{j} + 4\hat{k}$$

9. Find the angle between the vectors $2\hat{i}-\hat{j}+\hat{k}$ and $3\hat{i}+4\hat{j}-\hat{k}$.

Ans:

Given $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = 3\hat{i} + 4\hat{j} - \hat{k}$

Angle between vectors be θ then

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{1}{\sqrt{2^2 + 1^2 + 1^2} \sqrt{3^2 + 4^2 + 1^2}} (2\hat{i} - \hat{j} + \hat{k}) \cdot (3\hat{i} + 4\hat{j} - \hat{k})$$

$$\cos \theta = \frac{1}{\sqrt{6} \times \sqrt{26}} (6 - 4 - 1)$$

$$\cos \theta = \frac{1}{2\sqrt{39}}$$

$$\theta = \left(\frac{1}{2\sqrt{39}} \right)$$

10. If $\vec{a} + \vec{b} + \vec{c} = 0$, show that $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$. Interpret the result geometrically?

Ans:

$$\text{Given } \vec{a} + \vec{b} + \vec{c} = 0$$

$$\Rightarrow \vec{b} = -(\vec{a} + \vec{c})$$

$$\vec{a} \times \vec{b} = \vec{a} \times \{-(\vec{a} + \vec{c})\}$$

$$= -\vec{a} \times \vec{a} - \vec{a} \times \vec{c}$$

$$= 0 + \vec{c} \times \vec{a}$$

$$\vec{a} \times \vec{b} = \vec{c} \times \vec{a}$$

$$\text{Again, } \vec{b} \times \vec{c} = \{-(\vec{a} + \vec{c})\} \times \vec{c}$$

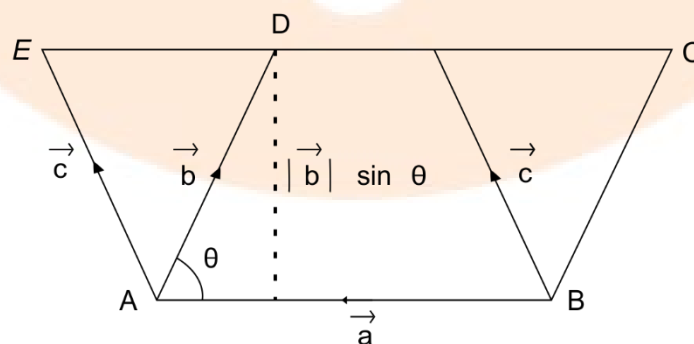
$$= -\vec{a} \times \vec{c} - \vec{c} \times \vec{c}$$

$$\vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

$$\text{Hence, } \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}.$$

Proved.

Geometrical interpretation of the result.



If ABCD is parallelogram such that $AB = \vec{a}$ and $AD = \vec{b}$ and these

adjacent sides are making angle θ between each other, then

$$\text{Area of parallelogram ABCD} = |\vec{a}| |\vec{b}| \sin \theta = |\vec{a} \times \vec{b}|$$

Since, parallelogram on the same base and between the same parallels are equal in area.

$$\text{So, } |\vec{a} \times \vec{b}| = |\vec{b} \times \vec{c}| = |\vec{c} \times \vec{a}|$$

This also implies that $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$

So, area of the parallelogram formed by taking any two sides represented

By $\vec{a}$, $\vec{b}$ and $\vec{c}$ as adjacent are equal.

11. Find the sine of the angle between the vectors $\vec{a}=3\hat{i}+\hat{j}+2\hat{k}$ and $\vec{b}=2\hat{i}-2\hat{j}+4\hat{k}$.

Ans:

Given that $\vec{a} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} - 2\hat{j} + 4\hat{k}$.

Let the angle between vectors be θ then

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ 2 & -2 & 4 \end{vmatrix} = (4+4)\hat{i} - (12-4)\hat{j} + (-6-2)\hat{k}$$

$$\vec{a} \times \vec{b} = 8\hat{i} - 8\hat{j} - 8\hat{k} \Rightarrow |\vec{a} \times \vec{b}| = 8\sqrt{1+1+1} = 8\sqrt{3}$$

$$|\vec{a}| = \sqrt{9+1+4} = \sqrt{14}$$

$$|\vec{b}| = \sqrt{4+4+16} = \sqrt{24}$$

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|} = \frac{8\sqrt{3}}{\sqrt{14}\sqrt{24}} = \frac{2 \times 4\sqrt{3}}{4\sqrt{3}\sqrt{7}}$$

$$= \frac{2}{\sqrt{7}}.$$

12. If A, B, C, D are the points with position vectors $\hat{i} + \hat{j} - \hat{k}$, $2\hat{i} - \hat{j} + 3\hat{k}$, $2\hat{i} - 3\hat{k}$, and $3\hat{i} - 2\hat{j} + \hat{k}$, respectively, find the projection of AB along CD.

Ans:

$$\text{Given, } OA = \hat{i} + \hat{j} - \hat{k}, OB = 2\hat{i} - \hat{j} + 3\hat{k},$$

$$OC = 2\hat{i} - 3\hat{k}, \text{ and } OD = 3\hat{i} - 2\hat{j} + \hat{k}$$

$$AB = OB - OA = (2\hat{i} - \hat{j} + 3\hat{k}) - (\hat{i} + \hat{j} - \hat{k}) = \hat{i} - 2\hat{j} + 4\hat{k}$$

$$CD = OD - OC = (3\hat{i} - 2\hat{j} + \hat{k}) - (2\hat{i} - 3\hat{k}) = \hat{i} - 2\hat{j} + 4\hat{k}$$

$$\text{Now, projection of AB along CD} = AB \cdot \frac{CD}{|CD|}$$

$$= \frac{1}{\sqrt{1^2 + 2^2 + 4^2}} (\hat{i} - 2\hat{j} + 4\hat{k}) \cdot (\hat{i} - 2\hat{j} + 4\hat{k})$$

$$= \frac{1}{\sqrt{21}} (1 + 4 + 16)$$

$$= \frac{21}{\sqrt{21}}$$

$$= \sqrt{21}$$

13. Using vectors, find the area of the triangle ABC with vertices

A (1, 2, 3), B (2, -1, 4) and C (4, 5, -1).

Ans:

Given A (1, 2, 3), B (2, -1, 4) and C (4, 5, -1).

$$AB = \hat{i} - 3\hat{j} + \hat{k}, \quad AC = 3\hat{i} + 3\hat{j} - 4\hat{k}$$

$$\text{Area of } \triangle ABC = \frac{1}{2} |AB \times AC|$$

$$= \frac{1}{2} \left\| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 1 \\ 3 & 3 & -4 \end{vmatrix} \right\|$$

$$= \frac{1}{2} \left\| (9\hat{i} + 7\hat{j} + 12\hat{k}) \right\|$$

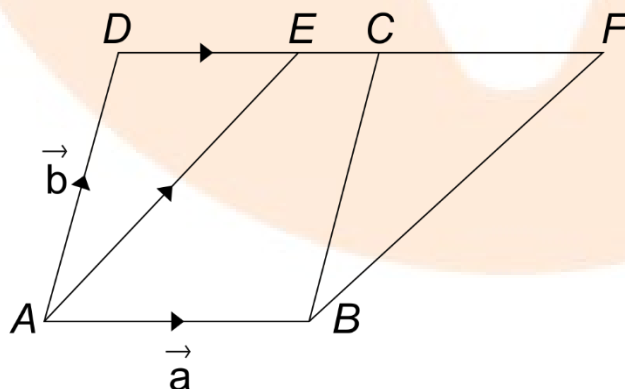
$$= \frac{1}{2} \sqrt{9^2 + 7^2 + 12^2}$$

$$= \frac{1}{2} \sqrt{81 + 49 + 144}$$

$$= \frac{1}{2} \sqrt{274} \text{ sq. units}$$

14. Using vectors, prove that the parallelogram on the same base and between the same parallels are equal in area.

Ans: Let ABCD and ABFE are parallelograms on the same base AB and between the same parallel line AB and DF.



Let $AB = \vec{a}$ and $AD = \vec{b}$

Area of parallelogram ABCD $= \vec{a} \times \vec{b}$

Again, area of parallelogram $ABFE = AB \times AE$

$$= AB \times (AD + DE)$$

$$= \vec{a} \times (\vec{b} + k\vec{a}); \because DE = k\vec{a}$$

$$= \vec{a} \times \vec{b} + 0$$

$$= \vec{a} \times \vec{b}$$

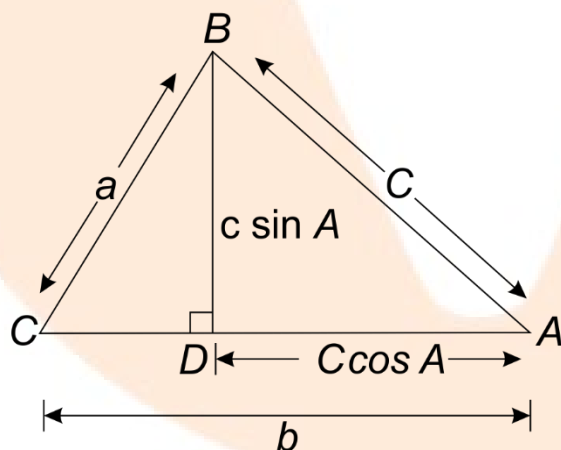
area of parallelogram $ABFE = \text{area of parallelogram } ABCD$

Hence, proved.

Long Answer (L.A.)

15. Prove that in any triangle ABC, $2bc \cos A = (b^2 + c^2 - a^2)$, where a, b, c are the magnitudes of the sides opposite to the vertices A, B, C, respectively.

Ans:



Here components of c are $c \cos A$ and $c \sin A$ is drawn.

Since, $CD = b - c \cos A$

In $\triangle BDC$, using Pythagoras theorem,

$$a^2 = (b - c \cos A)^2 + (c \sin A)^2$$

$$a^2 = b^2 + c^2 - 2bc \cos A + c^2 A$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

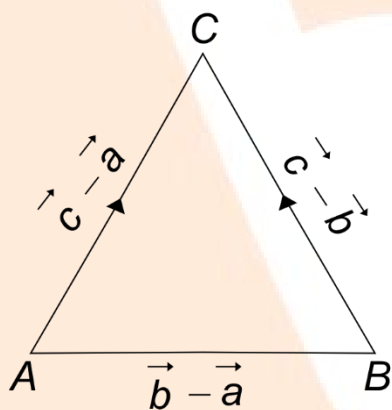
$$2bc \cos A = b^2 + c^2 - a^2$$

Hence proved.

16. If $\vec{a}, \vec{b}, \vec{c}$ determine the vertices of a triangle, show that $\frac{1}{2}[\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b}]$ gives the vector area of the triangle. Hence deduce the condition that the three points $\vec{a}, \vec{b}, \vec{c}$ are collinear. Also find the unit vector normal to the plane of the triangle.

Ans:

Given that $\vec{a}, \vec{b}, \vec{c}$ are the vertices of the $\triangle ABC$ as



$$\begin{aligned} \text{Area of } \triangle ABC &= \frac{1}{2} |\vec{AB} \times \vec{AC}| \\ &= \frac{1}{2} |(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})| \\ &= \frac{1}{2} |\vec{b} \times \vec{c} - \vec{b} \times \vec{a} - \vec{a} \times \vec{c} + \vec{a} \times \vec{a}| \\ &= \frac{1}{2} |\vec{b} \times \vec{c} + \vec{a} \times \vec{b} + \vec{c} \times \vec{a}| \end{aligned}$$

Hence proved.

For three points to be collinear $ar(\Delta ABC)$ will be zero. So,

$$\frac{1}{2} |\vec{b} \times \vec{c} + \vec{a} \times \vec{b} + \vec{c} \times \vec{a}| = 0$$

$$\vec{b} \times \vec{c} + \vec{a} \times \vec{b} + \vec{c} \times \vec{a} = 0$$

This is the required condition for collinearity of three points $\vec{a}, \vec{b}, \vec{c}$.

Let $\hat{n}$ be the unit vector normal to the plane of the ΔABC

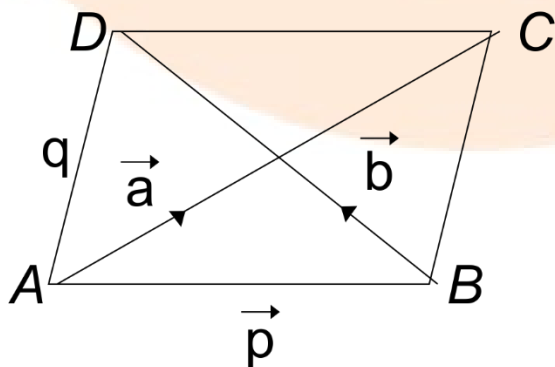
$$\therefore \hat{n} = \frac{AB \times AC}{|AB \times AC|}$$

$$= \frac{\vec{b} \times \vec{c} + \vec{a} \times \vec{b} + \vec{c} \times \vec{a}}{|\vec{b} \times \vec{c} + \vec{a} \times \vec{b} + \vec{c} \times \vec{a}|}$$

17. Show that area of the parallelogram whose diagonals are given by $\vec{a}$ and $\vec{b}$ is $\frac{|\vec{a} \times \vec{b}|}{2}$. Also find the area of the parallelogram whose diagonals are $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + 3\hat{j} - \hat{k}$.

Ans:

ABCD be a parallelogram such that



$$AB = \vec{p} = DC, AD = \vec{q} = BC$$

By triangle law of addition, we get

$$AC = \vec{p} + \vec{q} = \vec{a}$$

$$BD = -\vec{p} + \vec{q} = \vec{b}$$

$$AC + BD = 2\vec{q} = \vec{a} + \vec{b}$$

$$\vec{q} = \frac{1}{2}(\vec{a} + \vec{b})$$

$$\text{And } AC - BD = 2\vec{p} = \vec{a} - \vec{b}$$

$$\vec{p} = \frac{1}{2}(\vec{a} - \vec{b})$$

$$\text{Now, } \vec{p} \times \vec{q} = \frac{1}{4}(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b})$$

$$= \frac{1}{4}(\vec{a} \times \vec{a} + \vec{a} \times \vec{b} - \vec{b} \times \vec{a} - \vec{b} \times \vec{b})$$

$$= \frac{1}{4}(2\vec{a} \times \vec{b})$$

$$= \frac{1}{2}(\vec{a} \times \vec{b})$$

$$\text{Hence, area of parallelogram ABCD} = \vec{p} \times \vec{q} = \frac{1}{2}(\vec{a} \times \vec{b})$$

Area of parallelogram, whose diagonals are $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + 3\hat{j} - \hat{k}$

$$= \frac{1}{2}(2\hat{i} - \hat{j} + \hat{k}) \times (\hat{i} + 3\hat{j} - \hat{k})$$

$$= \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 1 & 3 & -1 \end{vmatrix}$$

$$= \frac{1}{2} \begin{vmatrix} -2\hat{i} + 3\hat{j} + 7\hat{k} \end{vmatrix}$$

$$= \frac{1}{2} \sqrt{2^2 + 3^2 + 7^2}$$

$$= \frac{1}{2} \sqrt{4 + 9 + 49}$$

$$= \frac{1}{2} \sqrt{62} \text{ sq. units}$$

18. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$, find a vector $\vec{c}$ such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$.

Ans: Given, $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$

Let, $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$

For, $\vec{a} \times \vec{c} = \vec{b}$,

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ x & y & z \end{vmatrix} = \hat{j} - \hat{k}$$

$$(z - y)\hat{i} - (z - x)\hat{j} + (y - x)\hat{k} = \hat{j} - \hat{k}$$

By comparing we have

$$z - y = 0 \Rightarrow y = z$$

$$z - x = -1 \Rightarrow x - z = 1$$

$$y - x = -1 \Rightarrow x - y = 1$$

Also, $\vec{a} \cdot \vec{c} = 3$

$$(\hat{i} + \hat{j} + \hat{k}) \cdot (x\hat{i} + y\hat{j} + z\hat{k}) = 3$$

$$x + y + z = 3$$

$$x + 2y = 3$$

By solving $x - y = 1$ and $x + 2y = 3$

So, $x = \frac{5}{3}, y = \frac{2}{3} = z$

$$\vec{c} = \frac{1}{3}(5\hat{i} + 2\hat{j} + 2\hat{k})$$

Objective Type Questions

Choose the correct answer from the given four options in each of the Exercises from 19 to 33 (M.C.Q)

19. The vector in the direction of the vector $\hat{i}-2\hat{j}+2\hat{k}$ that has magnitude 9 is

- (A) $\hat{i}-2\hat{j}+2\hat{k}$
- (B) $\frac{1}{3}(\hat{i}-2\hat{j}+2\hat{k})$
- (C) $3(\hat{i}-2\hat{j}+2\hat{k})$
- (D) $9(\hat{i}-2\hat{j}+2\hat{k})$

Ans:

Let $\vec{a} = \hat{i} - 2\hat{j} + 2\hat{k}$

Unit vector, $\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\hat{i} - 2\hat{j} + 2\hat{k}}{\sqrt{1^2 + 2^2 + 2^2}} = \frac{1}{3}(\hat{i} - 2\hat{j} + 2\hat{k})$

Required vector,

$$9 \frac{\vec{a}}{|\vec{a}|} = \frac{9}{3}(\hat{i} - 2\hat{j} + 2\hat{k}) = 3(\hat{i} - 2\hat{j} + 2\hat{k})$$

Option (c) is correct.

20. The position vector of the point which divides the join of points $2\vec{a}-3\vec{b}$ and $\vec{a}+\vec{b}$ in the ratio 3:1 is

(A) $\frac{1}{2}(3\vec{a}-2\vec{b})$

(B) $\frac{1}{4}(7\vec{a}-8\vec{b})$

(C) $\frac{1}{4}(3\vec{a})$

(D) $\frac{1}{4}(5\vec{a})$

Ans:

Let point P divides the join of points $2\vec{a}-3\vec{b}$ and $\vec{a}+\vec{b}$ in the ratio 3:1. So,

$$\begin{aligned}\text{Position vector of P} &= \frac{1}{3+1} \left[(2\vec{a}-3\vec{b}) + 3(\vec{a}+\vec{b}) \right] \\ &= \frac{1}{4}(5\vec{a})\end{aligned}$$

Option (D) is correct.

21. The vector having initial and terminal points as (2, 5, 0) and (-3, 7, 4), respectively is

(A) $-\hat{i}+12\hat{j}+4\hat{k}$

(B) $5\hat{i}+2\hat{j}-4\hat{k}$

(C) $-5\hat{i}+2\hat{j}+4\hat{k}$

(D) $\hat{i}+\hat{j}+\hat{k}$

Ans:

$$\begin{aligned}\text{Required vector} &= (-3-2)\hat{i} + (7-5)\hat{j} + (4-0)\hat{k} \\ &= -5\hat{i} + 2\hat{j} + 4\hat{k}\end{aligned}$$

Option (C) is correct.

22. The angle between two vectors $\vec{a}$ and $\vec{b}$ with magnitudes $\sqrt{3}$ and 4, respectively, and $\vec{a} \cdot \vec{b}$ is

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{2}$

(D) $\frac{5\pi}{2}$

Ans:

Given $|\vec{a}| = \sqrt{3}$, $|\vec{b}| = 4$ and $\vec{a} \cdot \vec{b} = 2\sqrt{3}$

$$\vec{a} \cdot \vec{b} = 2\sqrt{3}$$

$$|\vec{a}| |\vec{b}| \cos \theta = 2\sqrt{3}$$

$$\cos \theta = \frac{2\sqrt{3}}{4\sqrt{3}} = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

Option (B) is correct.

23. Find the value of λ such that the vectors, $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$ are orthogonal

(A) 0

(B) 1

(C) $\frac{3}{2}$

(D) $-\frac{5}{2}$

Ans:

Given nonzero vectors are $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$

If both the vectors are orthogonal then scalar product will be zero

$$\vec{a} \cdot \vec{b} = (2\hat{i} + \lambda\hat{j} + \hat{k}) \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 0$$

$$2 + 2\lambda + 3 = 0$$

$$\lambda = -\frac{5}{2}$$

Option (D) is correct.

24. The value of λ for which the vectors $3\hat{i} - 6\hat{j} + \hat{k}$ and $2\hat{i} - 4\hat{j} + \lambda\hat{k}$ are parallel is

(A) $\frac{2}{3}$

(B) $\frac{3}{2}$

(C) $\frac{5}{2}$

(D) $\frac{2}{5}$

Ans:

Given vectors are $3\hat{i} - 6\hat{j} + \hat{k}$ and $2\hat{i} - 4\hat{j} + \lambda\hat{k}$ are parallel then it may be written as

$$3\hat{i} - 6\hat{j} + \hat{k} = p(2\hat{i} - 4\hat{j} + \lambda\hat{k})$$

Where p is a nonzero scalar quantity.

By comparing we have

$$3 = 2p \Rightarrow p = \frac{3}{2}$$

$$-6 = -4p \Rightarrow p = \frac{3}{2}$$

$$1 = \lambda p \Rightarrow \lambda = \frac{1}{p} = \frac{2}{3}$$

Hence option (A) is correct.

25. The vectors from origin to the points A and B are $\vec{a} = 2\hat{i} - 3\hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} + 3\hat{j} + \hat{k}$, respectively, then the area of triangle OAB is

(A) 340

(B) 5

(C) $\sqrt{229}$

(D) $\frac{1}{2}\sqrt{229}$

Ans:

$$\begin{aligned} \text{Area of } \triangle ABC &= \vec{a} \times \vec{b} = \frac{1}{2} \left\| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 2 \\ 2 & 3 & 1 \end{vmatrix} \right\| \\ &= \frac{1}{2} \left| (-3-6)\hat{i} - (2-4)\hat{j} + (6+6)\hat{k} \right| \\ &= \frac{1}{2} \left| -9\hat{i} + 2\hat{j} + 12\hat{k} \right| \\ &= \frac{1}{2} \sqrt{81 + 4 + 144} \end{aligned}$$

$$\text{Area of } \triangle ABC = \frac{1}{2} \sqrt{229}$$

Option (D) is correct.

26. For any vector $\vec{a}$, the value of $(\vec{a} \times \hat{i})^2 + (\vec{a} \times \hat{j})^2 + (\vec{a} \times \hat{k})^2$ is equal to

(A) $\vec{a}^2$

(B) $3\vec{a}^2$

(C) $4\vec{a}^2$

(D) $2\vec{a}^2$

Ans: Let $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\vec{a} \times \hat{i} = (x\hat{i} + y\hat{j} + z\hat{k}) \times \hat{i}$$

$$= 0 - y\hat{k} + z\hat{j}$$

$$\vec{a} \times \hat{i} = -y\hat{k} + z\hat{j}$$

$$(\vec{a} \times \hat{i})^2 = (-y\hat{k} + z\hat{j})^2 = y^2 + z^2$$

Similarly, $(\vec{a} \times \hat{j})^2 = x^2 + z^2$ and $(\vec{a} \times \hat{k})^2 = x^2 + y^2$

Now, by adding we have

$$(\vec{a} \times \hat{i})^2 + (\vec{a} \times \hat{j})^2 + (\vec{a} \times \hat{k})^2 = 2(x^2 + y^2 + z^2)$$

$$\vec{a}^2 = x^2 + y^2 + z^2$$

$$(\vec{a} \times \hat{i})^2 + (\vec{a} \times \hat{j})^2 + (\vec{a} \times \hat{k})^2 = 2\vec{a}^2$$

Option (D) is correct.

27. If $|\vec{a}|=10$, $|\vec{b}|=2$ and $\vec{a} \cdot \vec{b}=12$, then value of $|\vec{a} \times \vec{b}|$ is

(A) 5

(B) 10

(C) 14

(D) 16

Ans: Given that $|\vec{a}| = 10$, $|\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = 12$

$$\vec{a} \cdot \vec{b} = 12$$

$$|\vec{a}| |\vec{b}| \cos \theta = 12$$

$$10 \times 2 \cos \theta = 12$$

$$\cos \theta = \frac{12}{20} = \frac{3}{5}$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - \left(\frac{3}{5}\right)^2}$$

$$\sin \theta = \frac{\sqrt{25 - 9}}{5} = \frac{4}{5}$$

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

$$= 10 \times 2 \times \frac{4}{5}$$

$$|\vec{a} \times \vec{b}| = 16$$

Option (D) is correct.

28. The vectors $\lambda \hat{i} + \hat{j} + 2\hat{k}$, $\hat{i} + \lambda \hat{j} - \hat{k}$ and $2\hat{i} - \hat{j} + \lambda \hat{k}$ are coplanar if (A) $\lambda = -2$

(B) $\lambda = 0$

(C) $\lambda = 1$

(D) $\lambda = -1$

Ans: Given vectors $\lambda \hat{i} + \hat{j} + 2\hat{k}$, $\hat{i} + \lambda \hat{j} - \hat{k}$ and $2\hat{i} - \hat{j} + \lambda \hat{k}$ are coplanar, so

$$\begin{vmatrix} \lambda & 1 & 2 \\ 1 & \lambda & -1 \\ 2 & -1 & \lambda \end{vmatrix} = 0$$

$$\lambda(\lambda^2 - 1) - (\lambda + 2) + 2(-1 - 2\lambda) = 0$$

$$\lambda^3 - \lambda - \lambda - 2 - 2 - 4\lambda = 0$$

$$\lambda^3 - 6\lambda - 4 = 0$$

$$(\lambda + 2)(\lambda^2 - 2\lambda - 2) = 0$$

$$\lambda = -2, \lambda = \frac{2 \pm \sqrt{12}}{2}$$

$$\lambda = -2, \lambda = \frac{2 \pm 2\sqrt{3}}{2}$$

$$\lambda = -2, \lambda = 1 \pm \sqrt{3}$$

Option (A) is correct.

29. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$, then the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ is

(A) 1

(B) 3

(C) $-\frac{3}{2}$

(D) None of these

Ans: Given that $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$

$$(\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) = 0$$

$$(\vec{a}^2 + \vec{b}^2 + \vec{c}^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})) = 0$$

$$1 + 1 + 1 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2}$$

Option (C) is correct.

30. Projection vector of $\vec{a}$ on $\vec{b}$ is

(A) $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$

(B) $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

(C) $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$

(D) $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \vec{a}$

Ans: Projection of $\vec{a}$ on $\vec{b}$ is $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

Projection vector of $\vec{a}$ on $\vec{b}$ is $\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \right) \frac{\vec{b}}{|\vec{b}|} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$

Option (A) is correct.

31. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$ and $|\vec{a}|=2, |\vec{b}|=3, |\vec{c}|=5$, then value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ is

(A) 0

(B) 1

(C) -19

(D) 38

Ans:

Given that $\vec{a}, \vec{b}, \vec{c}$ are vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$ and $|\vec{a}|=2, |\vec{b}|=3$ and $|\vec{c}|=5$,
,

$$(\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) = 0$$

$$(\vec{a}^2 + \vec{b}^2 + \vec{c}^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})) = 0$$

$$4 + 9 + 25 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{38}{2} = -19$$

Option (C) is correct.

32. If $|\vec{a}| = 4$ and $-3 \leq \lambda \leq 2$, then the range of $|\lambda \vec{a}|$ is

(A) $[0, 8]$

(B) $[-12, 8]$

(C) $[0, 12]$

(D) $[8, 12]$

Ans:

$$\text{Given } |\vec{a}| = 4 \text{ and } -3 \leq \lambda \leq 2$$

$$|\lambda \vec{a}| = |\lambda| |\vec{a}| = 4|\lambda|$$

$$\text{Now, } -3 \leq \lambda \leq 2 \Rightarrow 0 \leq |\lambda| \leq 3$$

$$0 \leq 4|\lambda| \leq 12$$

$$|\lambda \vec{a}| \in [0, 12]$$

Option (C) is correct.

33. The number of vectors of unit length perpendicular to the vectors $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{j} + \hat{k}$ is

(A) one

(B) two

(C) three

(D) infinite

Ans:

The number of vectors of unit length perpendicular to the vectors $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{j} + \hat{k}$ will be along the cross or vector product as

$$\hat{c} = \pm \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

There are two-unit vectors. Option (B) is correct.

Fill in the blanks in each of the Exercises from 34 to 40.

34. The vector $\vec{a} + \vec{b}$ bisects the angle between the non-collinear vectors $\vec{a}$ and $\vec{b}$ if _____

Ans: Given that vector $\vec{a} + \vec{b}$ bisects the angle between the non-collinear vectors $\vec{a}$ and $\vec{b}$ then let each angle is equal to θ

$$\text{Now, } \cos \theta = \frac{\vec{a} \cdot (\vec{a} + \vec{b})}{|\vec{a}| |\vec{a} + \vec{b}|} = \frac{\vec{b} \cdot (\vec{a} + \vec{b})}{|\vec{b}| |\vec{a} + \vec{b}|}$$

$$\frac{\vec{a} \cdot (\vec{a} + \vec{b})}{|\vec{a}|} = \frac{\vec{b} \cdot (\vec{a} + \vec{b})}{|\vec{b}|}$$

$$\frac{\vec{a} \cdot (\vec{a} + \vec{b})}{|\vec{a}|} - \frac{\vec{b} \cdot (\vec{a} + \vec{b})}{|\vec{b}|} = 0$$

$$\left(\frac{\vec{a}}{|\vec{a}|} - \frac{\vec{b}}{|\vec{b}|} \right) \cdot (\vec{a} + \vec{b}) = 0$$

35. If $\vec{r} \cdot \vec{a} = 0$, $\vec{r} \cdot \vec{b} = 0$ and $\vec{r} \cdot \vec{c} = 0$ for some non-zero vector $\vec{r}$, then the value of $\vec{a} \cdot (\vec{b} \times \vec{c})$ is _____

Ans:

Given that $\vec{r} \cdot \vec{a} = 0$, $\vec{r} \cdot \vec{b} = 0$ and $\vec{r} \cdot \vec{c} = 0$ also $\vec{r}$ is nonzero vector. So, $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar vectors. We know that scalar triple product of coplanar vectors is zero.

$$\therefore \vec{a} \cdot (\vec{b} \times \vec{c}) = 0$$

36. The vectors $\vec{a} = 3\hat{i} - 2\hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} - 2\hat{k}$ are the adjacent sides of a parallelogram. The acute angle between its diagonals is _____.

Ans: Given that $\vec{a} = 3\hat{i} - 2\hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} - 2\hat{k}$

Diagonals are

$$\vec{a} + \vec{b} = (3\hat{i} - 2\hat{j} + 2\hat{k}) + (-\hat{i} - 2\hat{k}) = 2\hat{i} - 2\hat{j}$$

$$\vec{a} - \vec{b} = (3\hat{i} - 2\hat{j} + 2\hat{k}) - (-\hat{i} - 2\hat{k}) = 4\hat{i} - 2\hat{j} + 4\hat{k}$$

Let acute angle between diagonals be θ then

$$\theta = \frac{(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b})}{|\vec{a} + \vec{b}| |\vec{a} - \vec{b}|} = \frac{(2\hat{i} - 2\hat{j}) \cdot (4\hat{i} - 2\hat{j} + 4\hat{k})}{|2\hat{i} - 2\hat{j}| |4\hat{i} - 2\hat{j} + 4\hat{k}|}$$

$$= \frac{8 + 4}{\sqrt{4 + 4} \sqrt{16 + 4 + 16}} = \frac{12}{\sqrt{8} \sqrt{36}}$$

$$= \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4}$$

37. The values of k for which $|k\vec{a}| < |\vec{a}|$ and $k\vec{a} + \frac{1}{2}\vec{a}$ is parallel to $\vec{a}$ holds true are _____.

Ans: Given that $|k\vec{a}| < |\vec{a}|$ and $k\vec{a} + \frac{1}{2}\vec{a}$

$$\Rightarrow |k| < 1 \Rightarrow -1 < k < 1$$

$$\text{Again, } k\vec{a} + \frac{1}{2}\vec{a} = \left(k + \frac{1}{2}\right)\vec{a}$$

We know that null vector has no specific direction so $k\vec{a} + \frac{1}{2}\vec{a}$ will be parallel to

$$\vec{a} \text{ when } k + \frac{1}{2} \neq 0 \Rightarrow k = -\frac{1}{2}$$

$$\text{Hence } k \in (-1, 1) - \left\{-\frac{1}{2}\right\}.$$

38. The value of the expression $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2$ is _____.

Ans:

$$\begin{aligned} |\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 &= |\vec{a}|^2 |\vec{b}|^2 \theta + |\vec{a}|^2 |\vec{b}|^2 \theta \\ &= |\vec{a}|^2 |\vec{b}|^2 (\theta + \theta) \\ &= |\vec{a}|^2 |\vec{b}|^2 \end{aligned}$$

39. If $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = 144$ and $|\vec{a}| = 4$, then $|\vec{b}|$ is equal to _____.

$$\begin{aligned} \text{Ans: } |\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 &= |\vec{a}|^2 |\vec{b}|^2 \theta + |\vec{a}|^2 |\vec{b}|^2 \theta \\ &= 4^2 |\vec{b}|^2 (\theta + \theta) \end{aligned}$$

$$144 = 16|\vec{b}|^2$$

$$|\vec{b}|^2 = 9$$

$$|\vec{b}| = 3$$

40. If $|\vec{a}|$ is any non-zero vector, then $(\vec{a} \cdot \hat{i})\hat{i} + (\vec{a} \cdot \hat{j})\hat{j} + (\vec{a} \cdot \hat{k})\hat{k}$ equals _____.

Ans: Let $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$

Now, $(\vec{a} \cdot \hat{i})\hat{i} = \{(x\hat{i} + y\hat{j} + z\hat{k}) \cdot \hat{i}\}\hat{i} = x\hat{i}$

Similarly, $(\vec{a} \cdot \hat{j})\hat{j} = y\hat{j}$ and $(\vec{a} \cdot \hat{k})\hat{k} = z\hat{k}$

$$(\vec{a} \cdot \hat{i})\hat{i} + (\vec{a} \cdot \hat{j})\hat{j} + (\vec{a} \cdot \hat{k})\hat{k} = x\hat{i} + y\hat{j} + z\hat{k} = \vec{a}$$

State True or False in each of the following Exercises.

41. If $|\vec{a}| = |\vec{b}|$, then necessarily it implies $\vec{a} = \pm\vec{b}$.

Ans: Given that $|\vec{a}| = |\vec{b}|$ then it doesn't imply that $\vec{a} = \pm\vec{b}$ because there are infinite number of vectors with same magnitude but different directions.

So, the given statement is False.

42. Position vector of a point P is a vector whose initial point is origin.

Ans: The given statement is True. Because the initial point of all position vectors is always the origin.

43. If $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, then the vectors $\vec{a}$ and $\vec{b}$ are orthogonal.

Ans: Given that $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$

$$\sqrt{a^2 + b^2 + 2\vec{a} \cdot \vec{b}} = \sqrt{a^2 + b^2 - 2\vec{a} \cdot \vec{b}}$$

$$a^2 + b^2 + 2\vec{a} \cdot \vec{b} = a^2 + b^2 - 2\vec{a} \cdot \vec{b}$$

$\vec{a} \cdot \vec{b} = 0$, if vectors are non-zero then both will be orthogonal otherwise not.

Here nothing is about the vectors whether they are non-zero or not.

Hence the given statement is False.

44. The formula $(\vec{a} + \vec{b})^2 = \vec{a}^2 + \vec{b}^2 + 2\vec{a} \times \vec{b}$ is valid for non-zero vectors $\vec{a}$ and $\vec{b}$.

$$\begin{aligned} \text{Ans: } (\vec{a} + \vec{b})^2 &= (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) \\ &= \vec{a}^2 + \vec{b}^2 + \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{b} \\ &= \vec{a}^2 + \vec{b}^2 + 2\vec{a} \cdot \vec{b} \end{aligned}$$

Hence the given statement is False.

45. If $\vec{a}$ and $\vec{b}$ are adjacent sides of a rhombus, then $\vec{a} \cdot \vec{b} = 0$.

Ans: Given that $\vec{a}$ and $\vec{b}$ are adjacent sides of a rhombus, so angle between them never be right angle

So, $\vec{a} \cdot \vec{b} \neq 0$

Hence the given statement is False.